

Lab 7 Batch G1

Dated 28-09-2026

1. An RR(1) parser is similar to an LL(1) parser except that it works from right-to left, tracing out a top-down, rightmost derivation. The parser continuously consults the rightmost symbol of the derivation and the rightmost symbol of the input string that has not yet been matched to make its decisions about what production to use. The construction and operation of this parser is analogous to the LL(1) case, except that instead of having FIRST and FOLLOW sets we have LAST and PRECEDE sets, where LAST(A) is the set of terminals that could come at the end of a production of A (plus ϵ if A can derive the empty string), and PRECEDE(A) is the set of the terminals that could come before a production of A in some sentential form. Similarly, instead of having an end-of-input marker, we have a beginning-of-input marker, which we denote \$. Write a program to implement RR(1) parsing.

Write a **C++ program to implement an RR(1) parser** for the following CFG:

$S \rightarrow A B C$

$S \rightarrow S A$

$A \rightarrow a b$

$A \rightarrow a$

$B \rightarrow b c$

$B \rightarrow b$

$C \rightarrow c d$

$C \rightarrow c$

2. Compute the **LAST set** for every non-terminal by processing the grammar from right to left.
3. Compute the **PRECEDE set** for every non-terminal.
4. Implement the RR(1) parsing algorithm.
5. Use \$ as the **beginning-of-input marker**.
6. Parse the input from **right to left**.
7. Report whether the input is **ACCEPTED** or **REJECTED**.

Test Inputs

abcd

abc

ababcd

ababc

abbcd

Expected Results

abcd → ACCEPTED

abc → ACCEPTED

ababcd → ACCEPTED

ababc → REJECTED

abbcd → REJECTED

Required Output

Display something similar to:

LAST sets:

$LAST(S) = \{ \dots \}$

$LAST(A) = \{ b, a \}$

$LAST(B) = \{ c, b \}$

$LAST(C) = \{ d, c \}$

PRECEDE sets:

$PRECEDE(S) = \{ \dots \}$

$PRECEDE(A) = \{ \dots \}$

$PRECEDE(B) = \{ \dots \}$

$PRECEDE(C) = \{ \dots \}$

Instructions

Due date: 28.09.2026, 11:30 A.M for Batch G1 (30 minutes extra is given for the documentation)

Cut-off date: 28.09.2026, 11:59 P.M. (Late submission deadline)

Note:

1. If the submission is after the due date/time, marks will be deducted for every 6 hours till the cut-off date. After the Cut-off date, No evolution and no marks
2. If the output is not shown for any of the above components, the mark cannot be given even if it is included in your documentation.